

MECH5200 Numerical Methods for PDEs : *Video 7:  
Finite Difference Expressions & Error Part I (Numerical  
Exploration)*

David Willis & Xinfang Jin

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# How to quantify the error in a meaningful manner?

- So far:
  - We have developed/determined several finite difference expressions for first and second derivatives
  - We have use finite differences to calculate the derivative of functions
  - We have use finite differences to solve a simple 1D PDE (really an ODE)
- **What we'd like to know:** How good are these approximations for solving PDEs?
  - Perhaps a better question: What does “good” even mean?
- Refer to MIT's OCW 16.920 Course Notes

# Start by looking at numerical error?

- Error = Actual Solution ( $u$ ) - Approximate Numerical Solution ( $\hat{u}$ ).

$$Error = u - \hat{u} \quad (1)$$

- How to “report this error”?
  - There is error at each node due to approximately satisfying the equation
  - Do we want to give error at each node, overall error, one number....?

# Start by looking at numerical error?

- Some ideas:
- We may want to know the *maximum error* in a given PDE solution (infinity norm):

$$\|Error\|_{\infty} = \max(|(u_i - \hat{u}_i)|) \quad (2)$$

- The error over the domain can be useful too – for this we can use a modified L2-norm to determine the error behavior:

$$\|Error\| = \left( \int [u(x) - \hat{u}(x)]^2 dx \right)^{1/2} \approx \sqrt{\sum_{i=1}^{i=N} ((u_i - \hat{u}_i)^2 \cdot \Delta x)} \quad (3)$$

- Let's see how  $\|Error\|$  improves as  $\Delta x \rightarrow 0$  numerically.

# How to quantify the error in a meaningful manner?

- Let's start with:

$$\nabla^2 u = 1 \tag{4}$$

- What does error do with respect to  $\Delta x$ ? Why?

# Solution and Error FD solution of $\nabla^2 u = 1$ , $N = 11$

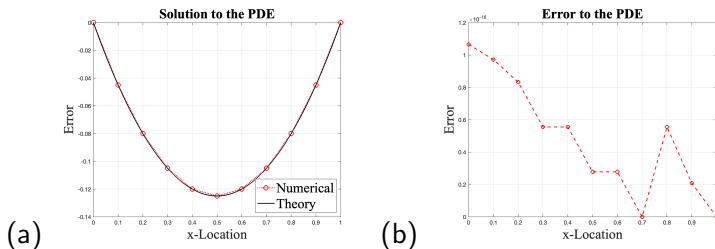


Figure: (left) Solution, (right) Error

- For this problem, the analytical solution is a quadratic polynomial.
  - 1 **Theory:** The finite difference equations capture the second derivative of a quadratic polynomial “exactly”
  - 2 **Numerically:** error is introduced by number representation in the computer, IEEE 754 Standard – double precision or FP64. See wiki:

| [https://en.wikipedia.org/wiki/Double-precision\\_floating-point\\_format](https://en.wikipedia.org/wiki/Double-precision_floating-point_format)

# Solution and Error FD solution of $\nabla^2 u = 1$ , $N = 101$

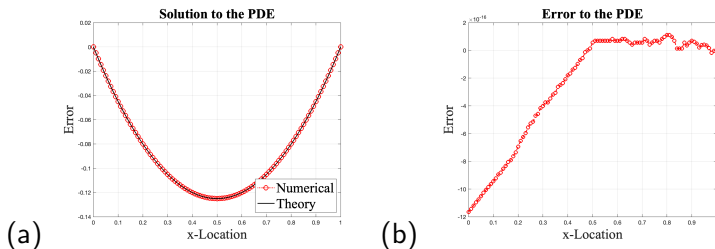


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# FD Error: Numerical Representation and Discretization Error

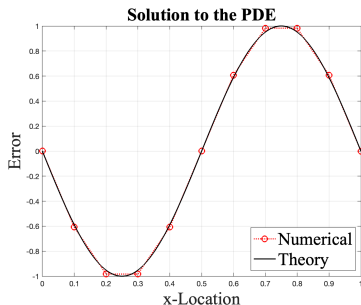
- In the previous example, the primary error was due to the finite representation of numbers in the computer.
- Let's make the ODE/PDE from earlier a bit more complicated – update the RHS:

$$\nabla^2 u = p = 4\pi^2 \sin(2\pi x) \quad (5)$$

- Now, how does the error behave? Does it change when we change  $\Delta x$ ? Why?

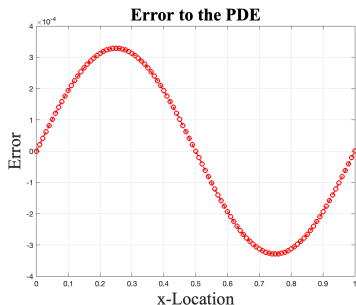
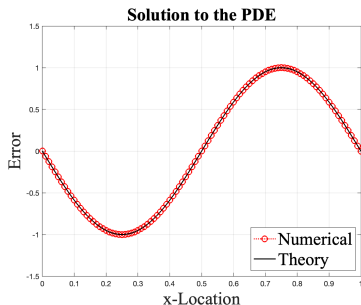


Solution:  $p = 4\pi^2 \sin(2\pi x)$ ,  $N = 11$



- We can see solution is reasonable (compared to expected solution)
- Error is now non-zero across most of  $x \in [0, 1]$ .
- Maximum error,  $\|e\|_{\infty} = 0.0319159...$
- L2 Modified error,  $\|e\|_2 = 0.0237293....$

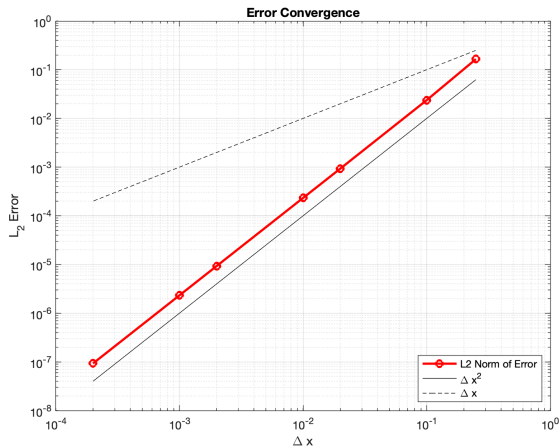
Solution:  $p = 4\pi^2 \sin(2\pi x)$ ,  $N = 101$



- More points  $\rightarrow$  “better” answer
- Error still non-zero across most of  $x \in [0, 1]$ .
- Maximum error,  $\|e\|_{\infty} = 3.290517629399581e - 04$
- L2 Modified error,  $\|e\|_2 = 2.326747329331382e - 04$

# Solution

- The modified L2 Norm error convergence – we can plot error vs.  $\Delta x$  on a log-log plot to observe convergence rate:



# What have we learned?

- Error decreases as we increase the number of nodes in the numerical solution
- If the approximation is second order and the solution is constant, linear, or quadratic, we solve the problem exactly.
- Next, we will determine whether we can theoretically show error behavior.